

Question number	Answer	Additional guidance	Mark
8(a)	A diagram that shows: • COOH group and NH₂ group (1) • H and the aspartate R group (1) • all attached to a central C (1)	Accept charged groups	(3)

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8(b)	An explanation that includes the following points: • because (the urea cycle has) many {stages / steps / reactions} (1) • therefore, the product of one stage is the substrate of the next stage (1) • and enzymes are specific to one substrate (1)	ACCEPT each stage has {new / different} substrate / different substrates (in the process)	(3)

	substrates {bind / attach / fit} to (specific) active site / credit reference to lock and key theory / credit reference to induced fit (1)	ACCEPT if active site is referred to in the context of an enzyme-substrate complex	
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Question number	Answer	Additional guidance	Mark
8(c)(i)	An answer that includes three of the following points: - genetic screening / named screening method / looking for a mutation - biochemical test / blood test / description of named molecule whose level would be different - family history / pedigree analysis	IGNORE where the molecules maybe found	(3)

Question number	Answer
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*8(c)(ii)	Indicative content: Table of results: • OTC mRNA treatment reduces the levels of ammonia back to nearly normal (Diagram of urea cycle: • ammonia levels will build up if carbamyl phosphate is not combined with ornithine • OTC deficiency will result in carbamyl phosphate not combining with ornithine Diagram of particle: • particle targets liver and this is where the urea cycle takes place • the phospholipids will fuse with the liver cells to enable the mRNA to enter the cells • because the phospholipids can move • the phospholipids will protect the mRNA • from hydrolytic enzymes • the mRNA will be translated inside the liver cells • producing functional OTC • the OTC will combine carbamyl phosphate with ornithine • reducing the levels of urea
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Aspects to cover :

D_d - description of data

E_d - explanation of data : ammonia converted to urea because ornithine cycle functioning

D_r - recognition of mRNA involved in translation

E_r - explanation that functional OTC can be produced

D_p - comment on role of particle

E_p - explanation of phospholipid particle

Level 1

1 mark : one description

2 marks : one description and one explanation or two descriptions

Level 2

3 marks : two descriptions and one explanation or three descriptions

4 marks : two descriptions and two explanations

Level 3

5 marks : three descriptions and two explanations

	6 marks : three descriptions and three explanations
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